Structured Probabilistic Models

for Hierarchical Clustering of Individual Stocks or Market Indices

Team Members

Anusha Shrestha

Bivash Yadav

Suprabha Pudasainy

Susta Pakhreen

Yuva Raj Bhatta

Deliverable 3: Poster Outlining the Process of the Research Paper
Dr. April Adams | Spring 2026
University of the Cumberlands
MSDS-690-M40- Data Science Capstone: Theory

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Background of the Topic

1. Financial markets exhibits hierarchical structures where asset correlations reflect economic relationships
2. Correlation-based networks reveal hidden dependencies among financial assets
3. Financial data is noisy; probabilistic filtering improves reliability of detected relationships
4. Hierarchical clustering captures multi-level organization such as sectors and sub-sectors
5. Graph-based approaches are widely used in modern financial data science

Methodology & Design

Methodology

Structured Probabilistic Modeling

Analyzes correlation structures through statistical filtering to identify reliable asset relationships within financial networks.

Design Approach

Network-Based Hierarchical Clustering

Builds MST networks from correlation matrices, applies bootstrap reliability testing, and clusters assets without predefined groupings.

Analysis Pipeline

Price Data Collection → Log Return Transformation → Correlation Matrix → MST Filtering & Bootstrap → Hierarchical Clustering → Temporal Analysis

Source: Bonanno et al. (2003); Tumminello et al. (2007)

Overarching Question & Sub-Questions

Can structured probabilistic modeling and hierarchical clustering identify stable and economically meaningful structures in financial markets?

Sub-Questions

01 Do the identified clusters correspond to real economic sectors or industries?

02 How stable are these clusters over time across different market conditions?

03 How does market structure change during periods of financial stress or crisis?

04 Can probabilistic filtering improve the reliability of detected relationships?

Conclusion

1. Financial markets exhibit hierarchical structures effectively captured through correlation-based clustering, revealing economically meaningful asset groupings.

2. Probabilistic filtering techniques such as Minimum Spanning Trees improve financial network reliability by reducing noise and highlighting significant relationships.

3. Hierarchical clustering combined with temporal analysis provides insight into evolving market dynamics, supporting diversification and systemic risk assessment.

Summary

Financial markets exhibit structured relationships identifiable via correlation-based clustering techniques.

Probabilistic filtering improves reliability by reducing noise and highlighting meaningful asset connections.

Hierarchical clustering reveals multi-level market structures useful for diversification and risk analysis.

Rolling window analysis captures temporal changes in market structure across different economic conditions.

Visualization techniques like heatmaps, MST graphs, dendrograms, improve interpretation of complex financial data.

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Group Short Bios

Anusha Shrestha

Currently a full time Data Science student. In my spare time, I like to explore new places, go for hikes or watch movies.

Susta Pakhreen

Currently working as a Data Science Intern. I enjoy hiking, practicing Yoga, exploring new places and spending quality time with friends and family.

Bivash Yadav

I am Software Developer. In my spare time, I like to drive around scenic roads or go far a hike.

Yuva Raj Bhatta

Been working as a full stack java developer in payroll-based company. I love to sing and play instruments. Also, love to roam around in a nature.

Suprabha Pudasainy

I'm Software Developer working in the health insurance industry. In my spare time, I like to read, hike, and cook.

Thank You

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